

Better Approximation for Weighted k -Matroid Intersection

Neta Singer

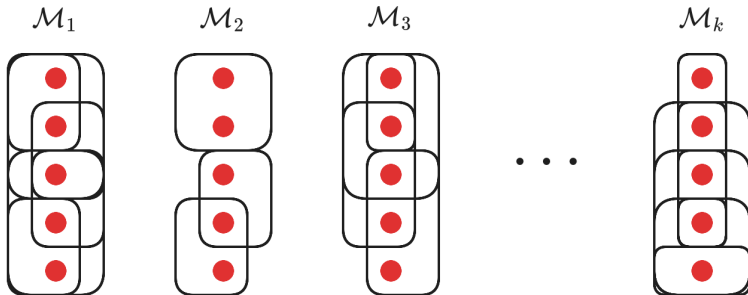
École Polytechnique Fédérale de Lausanne

Joint work with Theophile Thiery

June 25, 2025

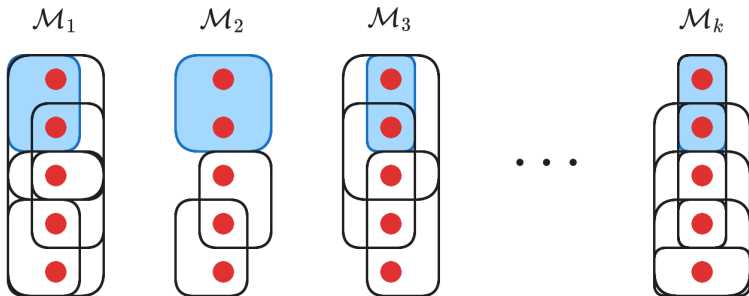
The matroid intersection problem

- **Input:** k matroids over E , weight function $w : E \rightarrow \mathbb{R}_{\geq 0}$



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- **Goal:** find set $S \subseteq E$ s.t. $S \in \bigcap_{i=1}^k \mathcal{M}_i$ and $w(S)$ is maximized.



The matroid intersection problem: motivation

- **2 matroids:**
 - captures bipartite matching, arborescences, and more
 - poly time solvable (hungarian method)

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- **3+ matroids:**

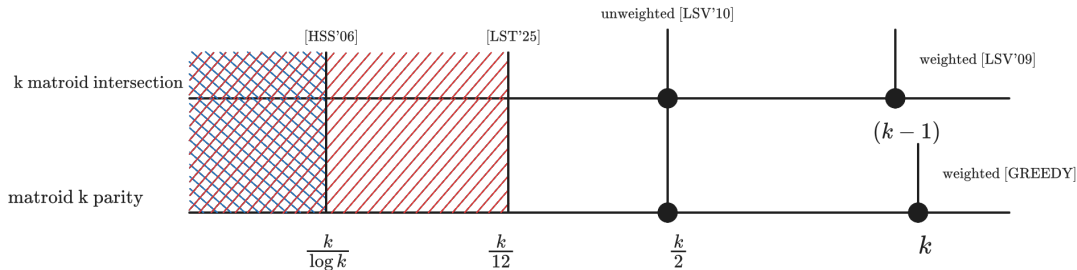
- captures matching in k -partite graphs, path packing, minimum pinning set, maximum genus embedding, maximum planar subgraph and more
- NP-hard

Prior guarantees

- Approximation algorithm where output A satisfies $w(OPT) \leq \alpha \cdot w(A)$.

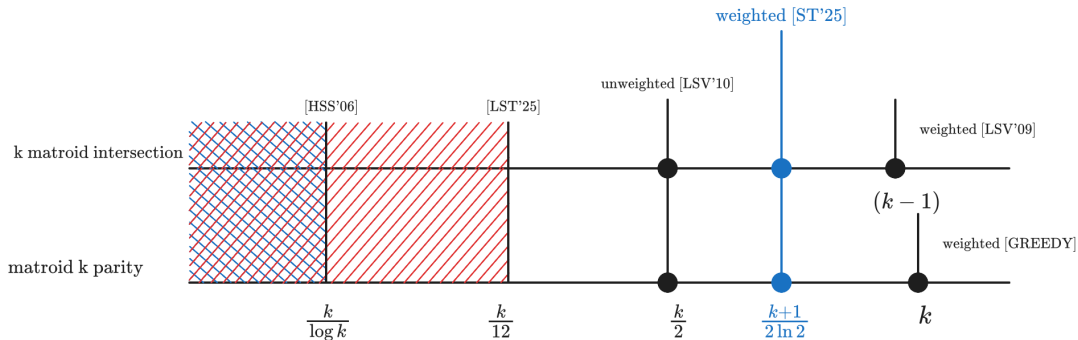
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Our results

Theorem (S., Thiery 2025)

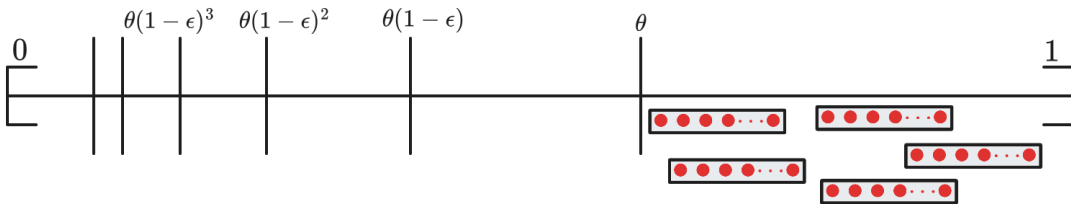
*There exists a $\frac{k+1}{2\ln 2}$ -approximation algorithm for **weighted** k -Matroid Intersection.*

Theorem (S., Thiery 2025)

*There exists a $\frac{k+1}{2\ln 2}$ -approximation algorithm for **weighted** Matroid k -Parity.*

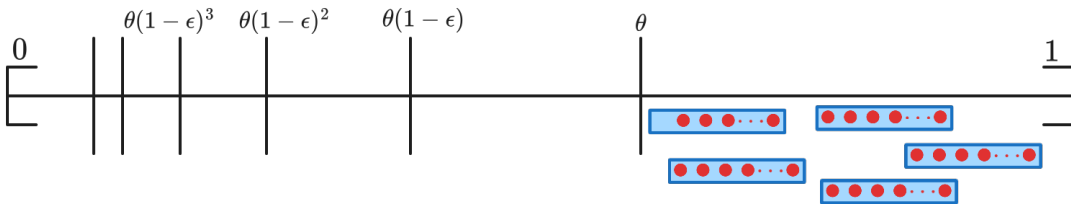
Our algorithm

- Let $\epsilon \in (0, 1)$ a well chosen parameter.
- Sample $\theta \sim \mathcal{U}[1 - \epsilon, 1]$.
- Construct geometric weight scale and bucket items by their weight class.



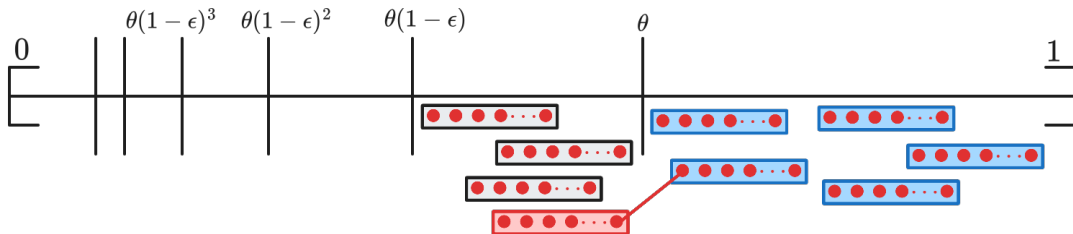
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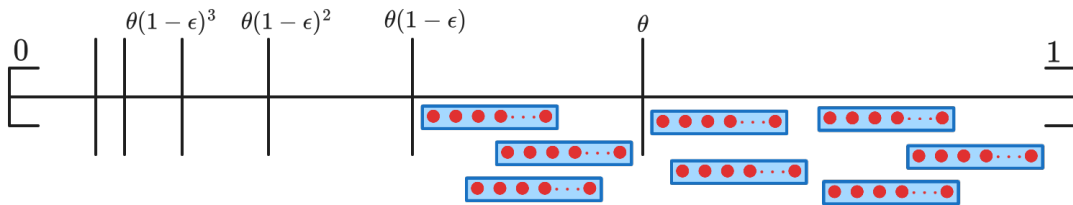
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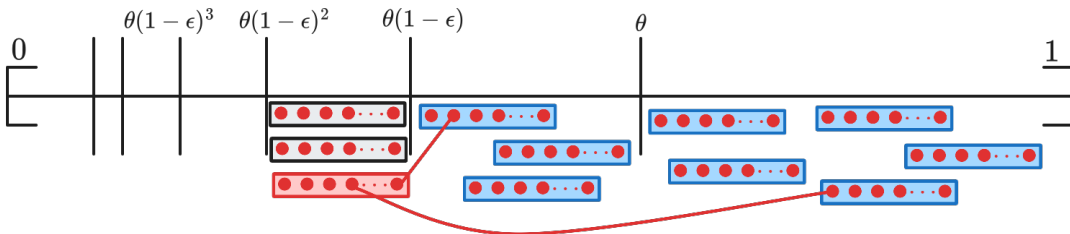
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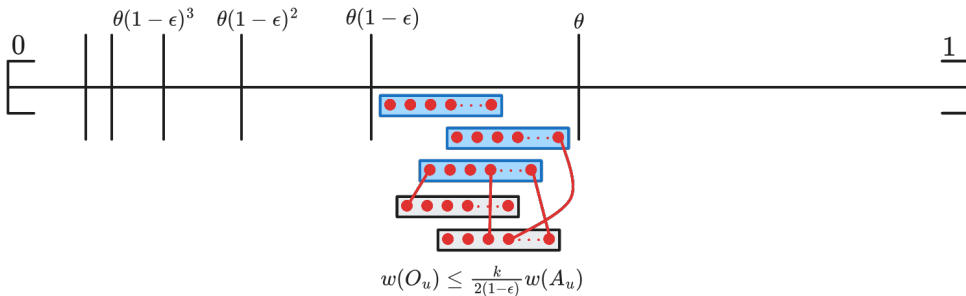


- If $A \setminus \{a\} \cup \{o\}$ is feasible in some matroid, we say o is *blocked* by a .



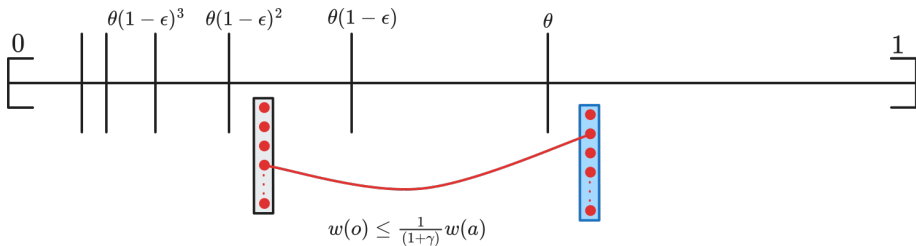
Algorithm analysis: intuition

- Elements not *blocked* by decisions from higher weight buckets behave like an unweighted instance.



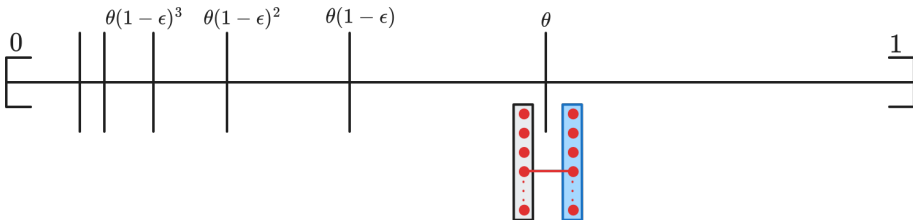
Algorithm analysis: intuition

- Elements of OPT blocked by a much higher weight item are *good*.



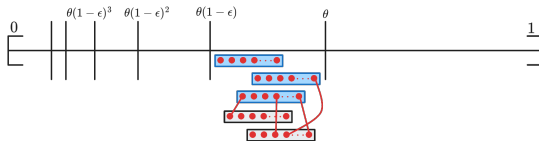
Algorithm analysis: intuition

- Elements of OPT blocked by a prior decision of almost same weight are *bad*.



Algorithm analysis: intuition

- **Unweighted items O_u :**
 $w(O_u) \leq \left(\frac{k}{2(1-\epsilon)} + 1 \right) w(A)$

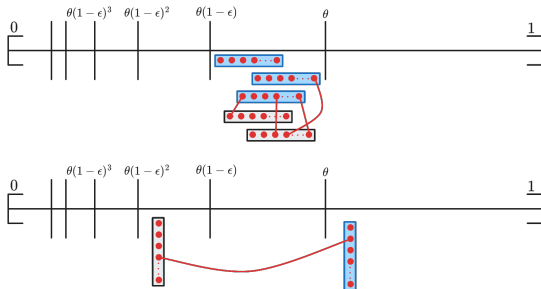


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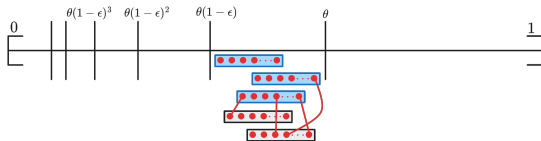
$$w(O_u) \leq \left(\frac{k}{2(1-\epsilon)} + 1 \right) w(A)$$
- **Good items O_g :**

$$w(O_g) \leq \frac{k}{(1+\gamma)} w(A)$$

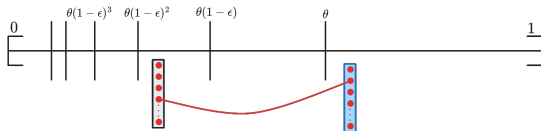


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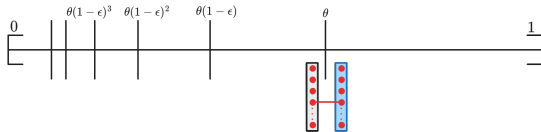
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- **Bad items O_b :**
 $\mathbb{P}[o \in O_b] \leq \frac{\gamma}{\epsilon(1+\gamma)}$



Our Results

Lemma (S., Thiery 2025)

After discarding the smallest weight interval,

$$\mathbb{E}[w(O_g)] + \mathbb{E}[w(O_s)] + \mathbb{E}[w(O_d)] \leq \left(\frac{k}{1+\gamma} + 1 \right) \mathbb{E}[w(A)]$$

for some well chosen $\gamma \in (0, 1)$.

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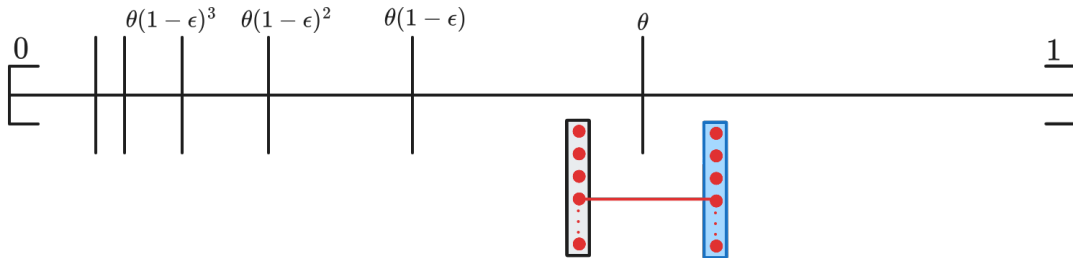
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Theorem (S., Thiery 2025)

There is $\frac{9(k+1)}{10}$ -approximation algorithm for weighted k -Matroid Intersection.

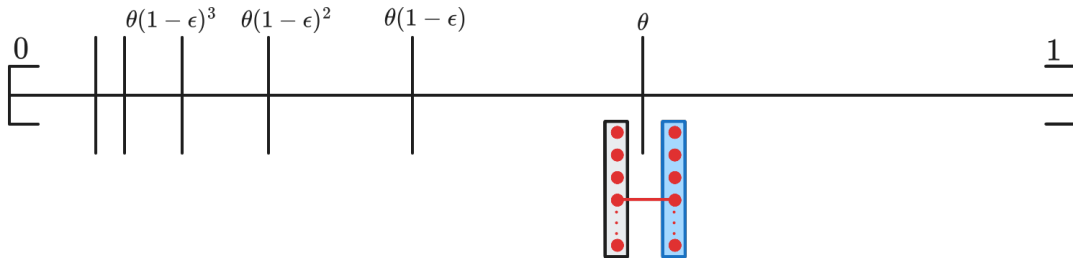
Our Results

- Using a tighter analysis, can bring the constant factor down.



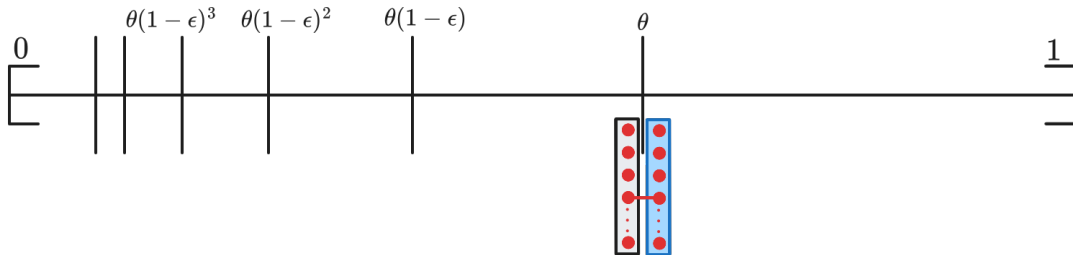
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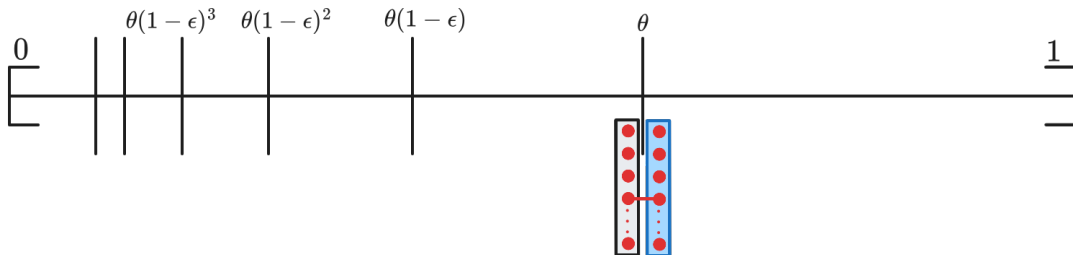
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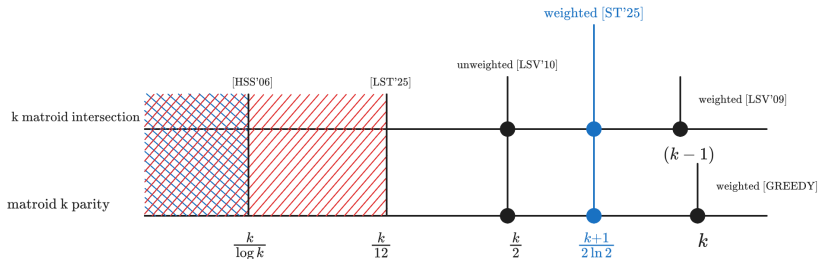


Theorem (S., Thiery 2025)

There is $\frac{k+1}{2\ln 2}$ -approximation algorithm for weighted k -Matroid Intersection.

Future directions

- Achieve best approximation guarantee.



- Extend idea to other independence systems.
- Extend idea to other maximization frameworks.

Thank you!

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